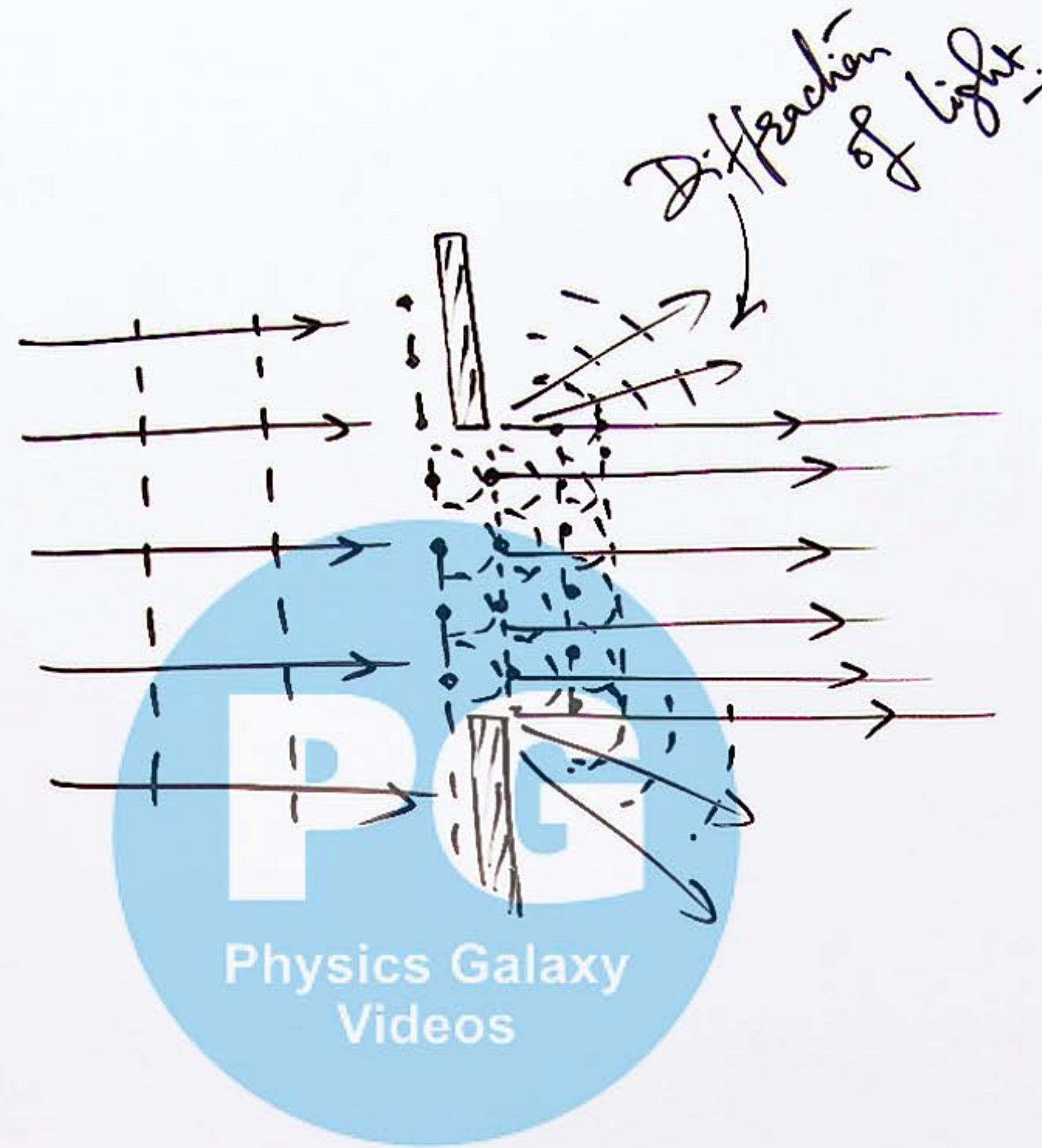


① Diffraction of Light :

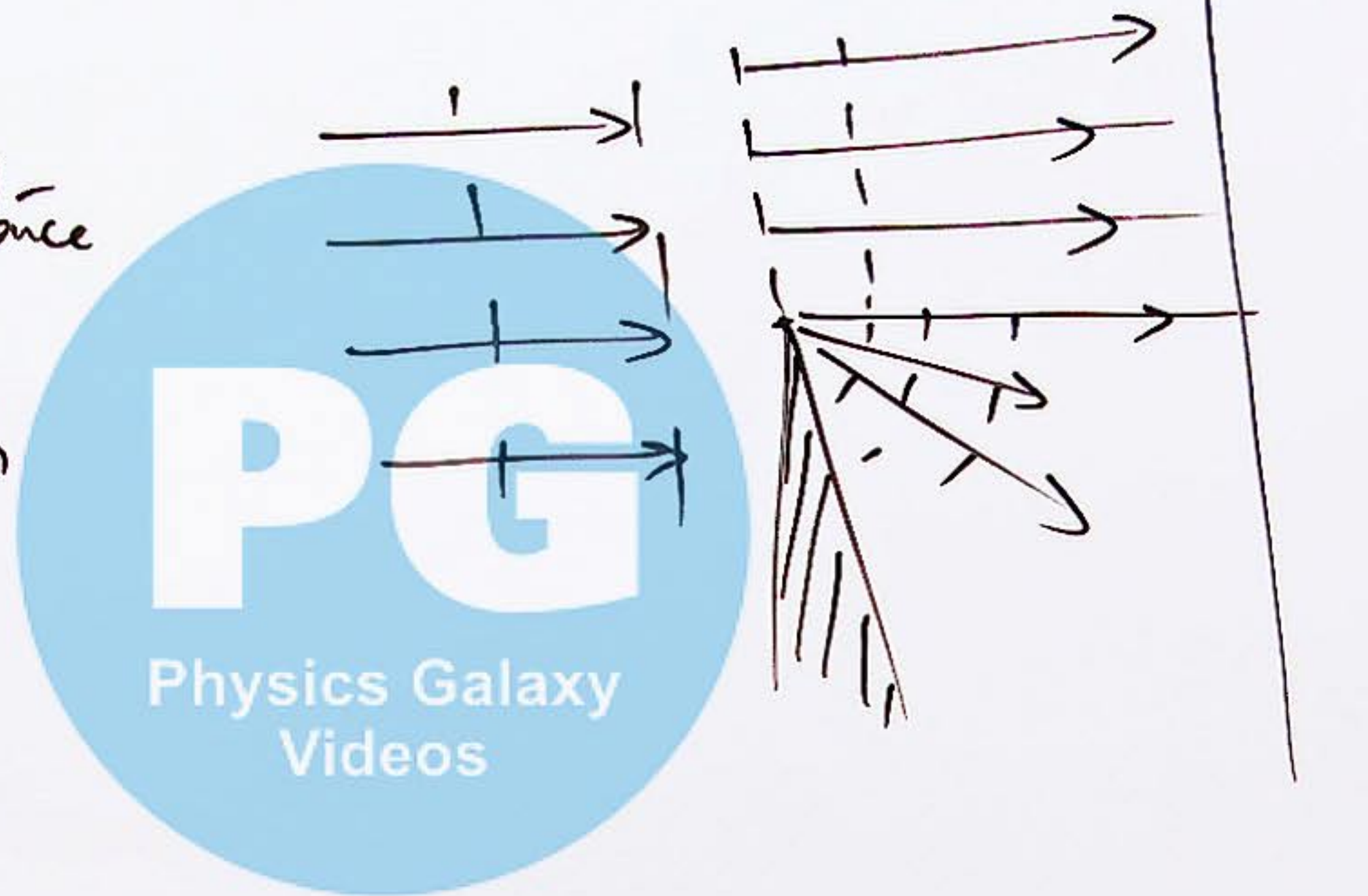


② Types of Diffraction:

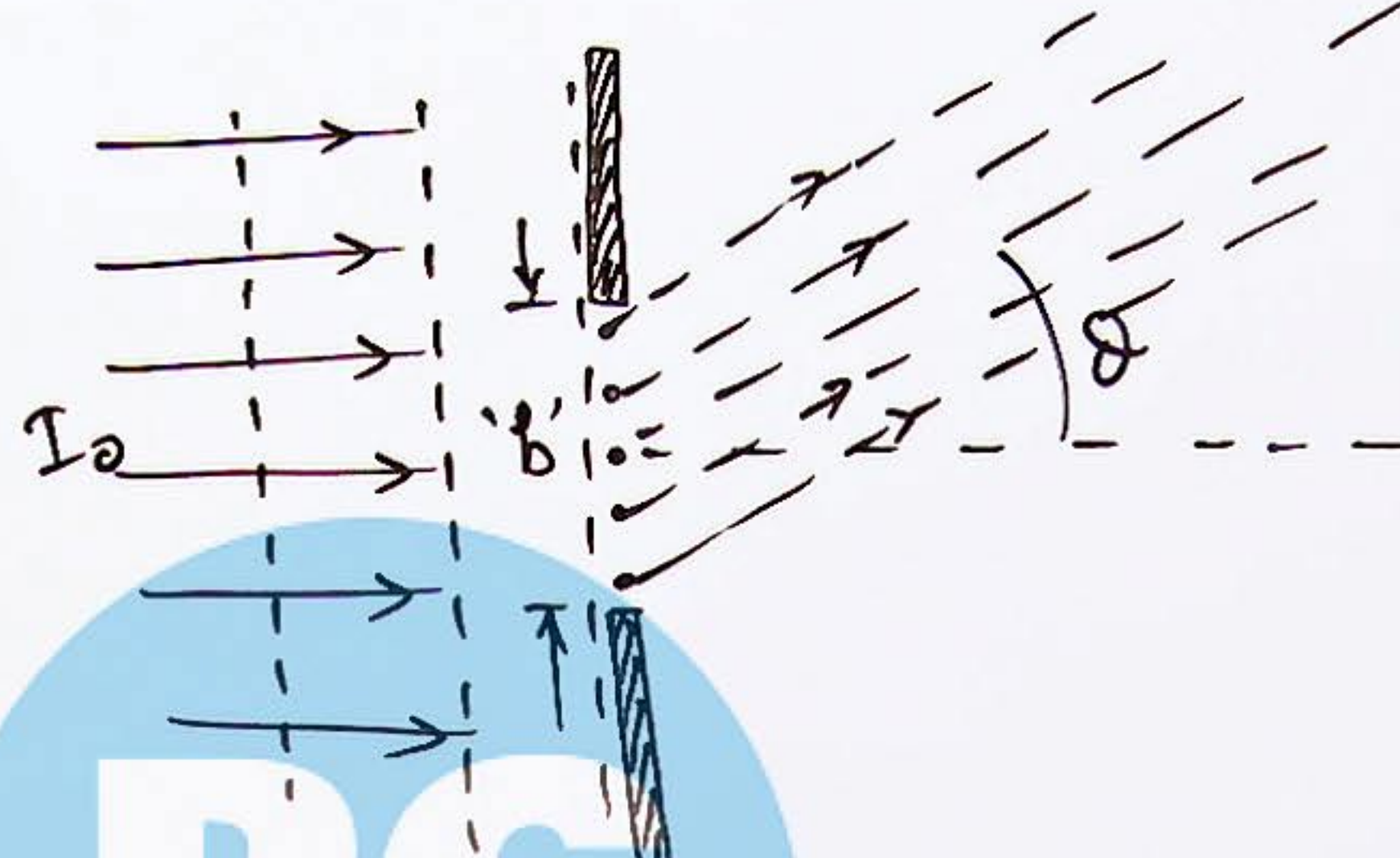
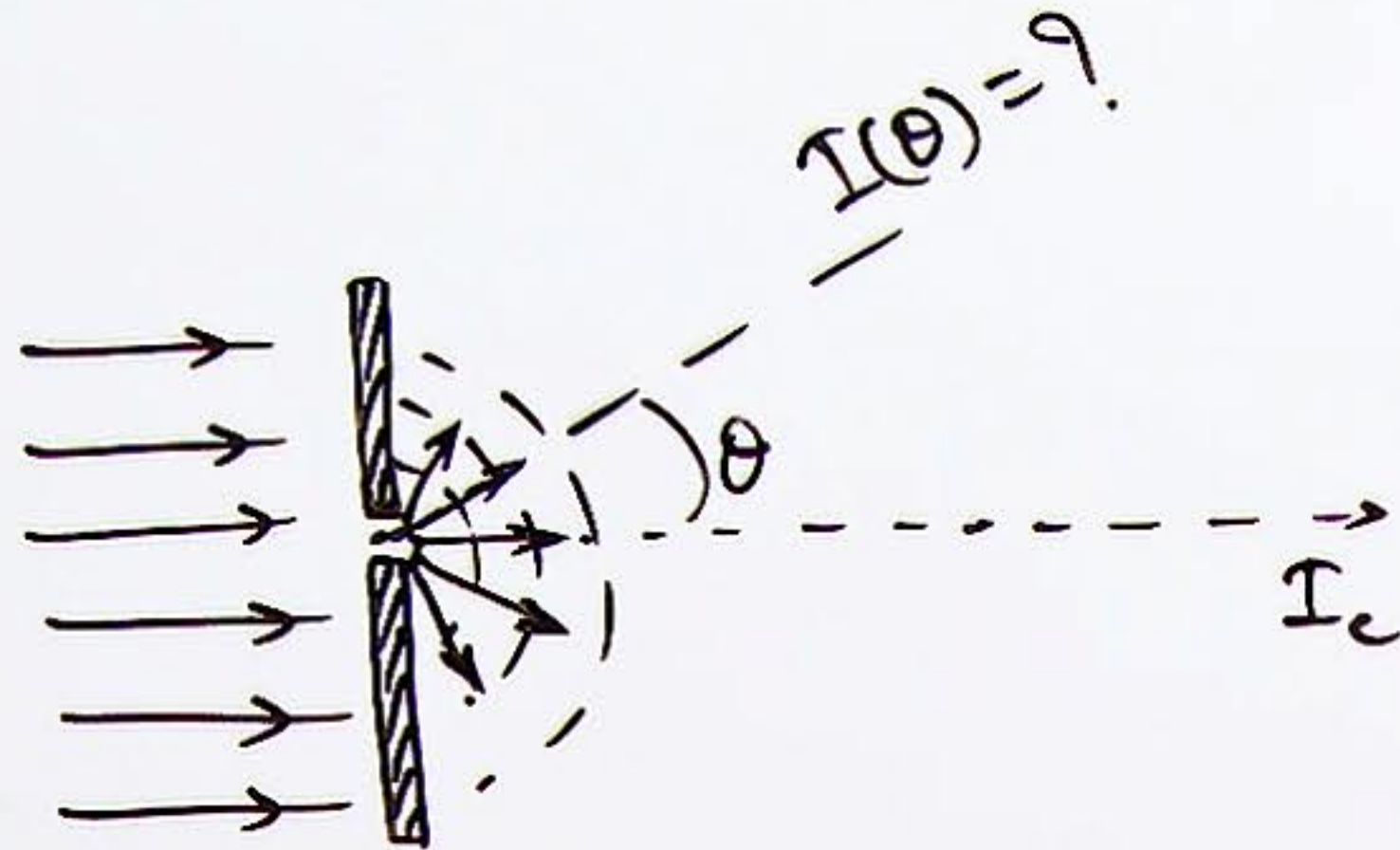
(i) Fresnel Diffraction $\rightarrow$ for finite distance observation

(ii) Fraunhofer Diffraction

$\downarrow$
for v. far infinite distance
observation



③ Diffraction by a Single Slit :



$$I_{\theta} = I_0 \frac{\sin^2 \beta}{\beta^2}$$

where

$$\beta = \frac{\pi b \sin \theta}{\lambda}$$

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④ Diffraction minima due to Single Slit :

if $\theta = 0 \Rightarrow \beta = 0$

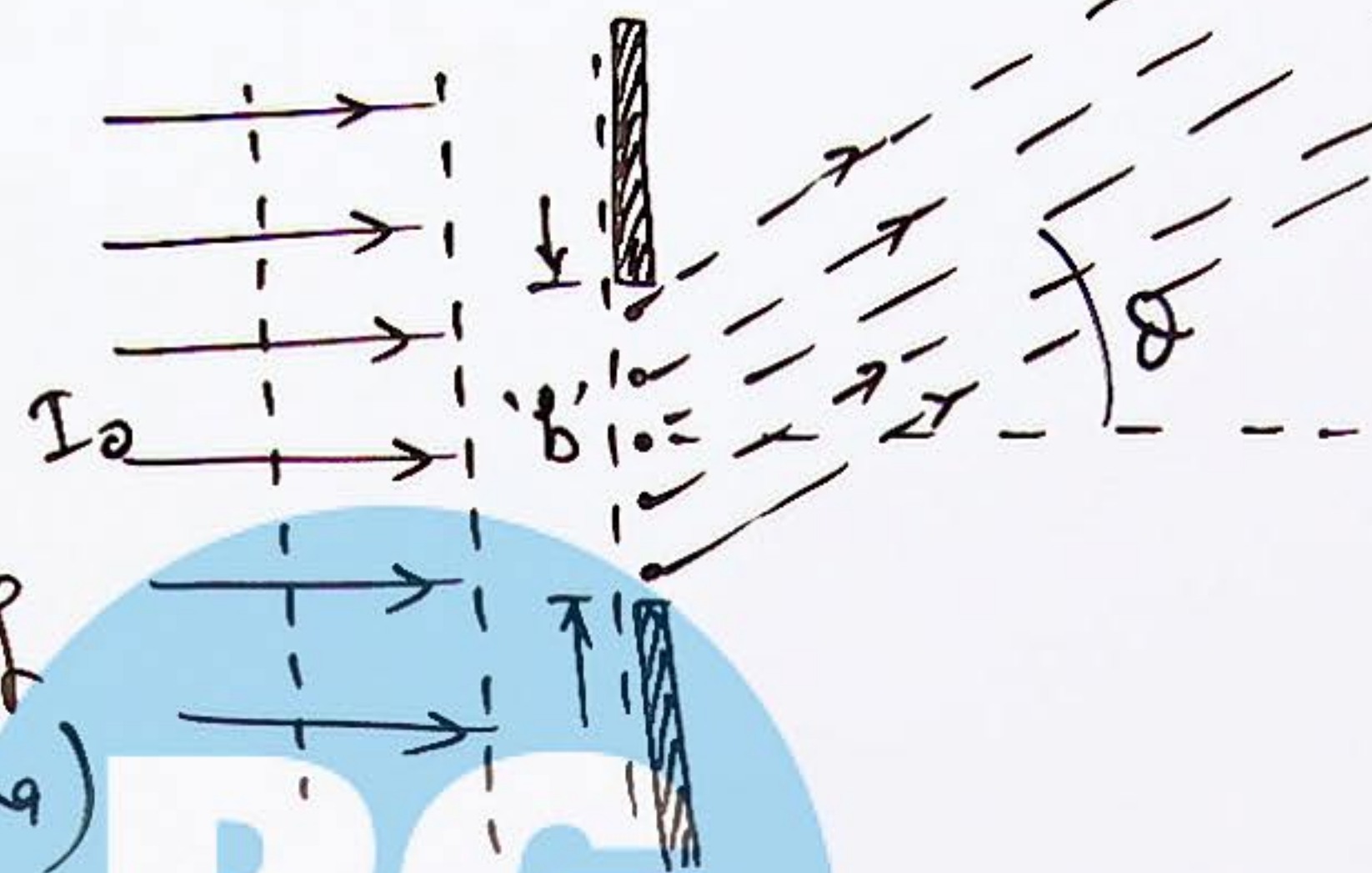
lit $\lim_{\beta \rightarrow 0} \frac{\sin \beta}{\beta} = 1$

$\Rightarrow I_{\theta=0} = I_0$

if $\beta = m\pi \Rightarrow I_{\theta} = 0$ (Position of Minima)

$m \neq 0$ $m\pi = \frac{\pi b \sin \theta}{\lambda}$

$\theta = \sin^{-1} \left(\frac{m\lambda}{b} \right)$



$I_{\theta} = I_0 \frac{\sin^2 \beta}{\beta^2}$

where

$\beta = \frac{\pi b \sin \theta}{\lambda}$

line of Central maxima.

first minima is obtained at Angular.

$\theta_1 = \sin^{-1} \frac{\lambda}{b} \approx \frac{\lambda}{b}$

Central fringe width is

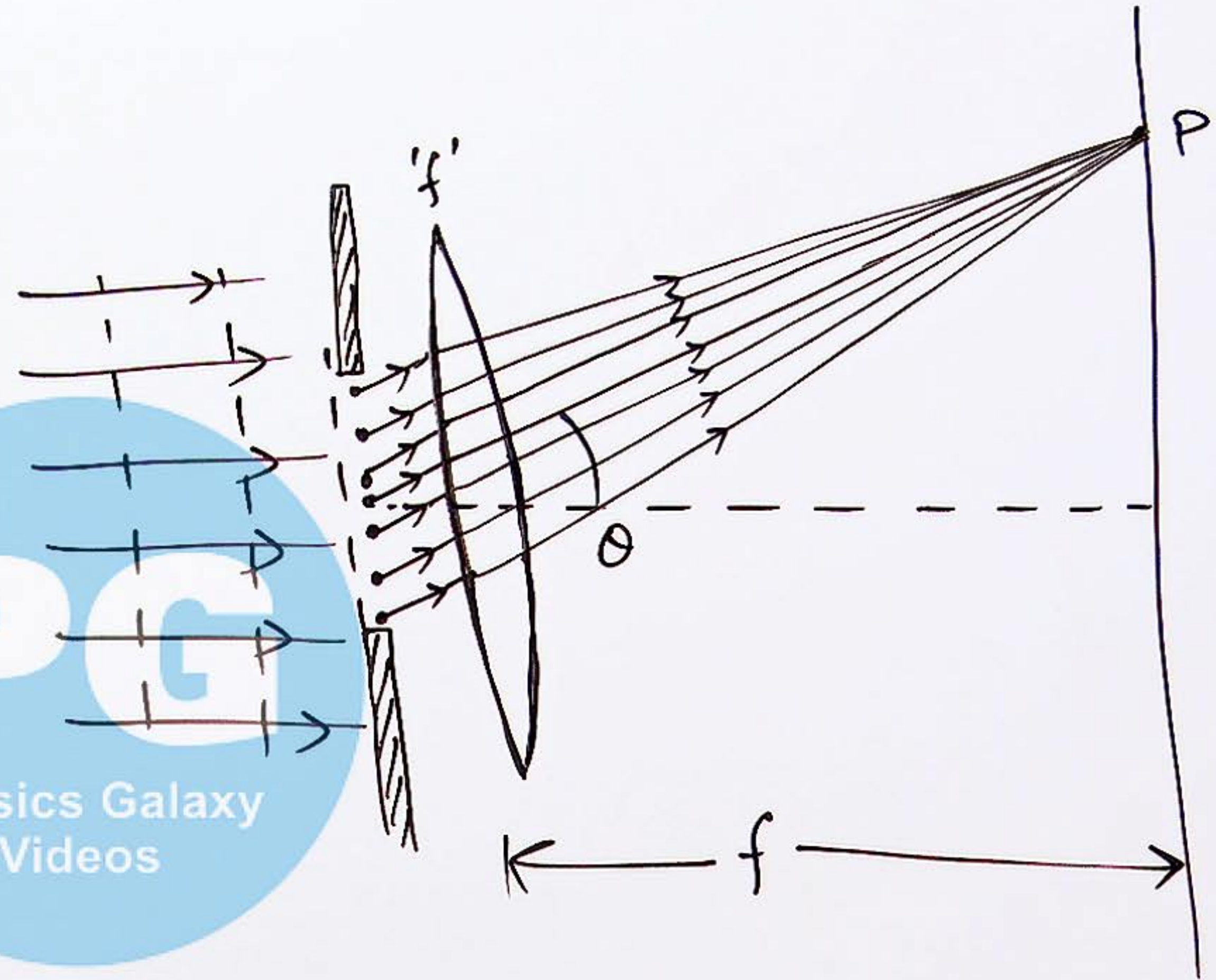
$\delta = 2\theta_1 = \frac{2\lambda}{b}$

⑤ Fringe pattern in diffraction
due to single slit :

$$I_{\theta} = I_0 \frac{\sin^2 \beta}{\beta^2}$$

Angular positions of minima

$$\sin \theta = \frac{m\lambda}{b} \quad (m \neq 0)$$



⑥ Diffraction Maxima due to Single Slit :

$$I(\theta) = I_0 \frac{\sin^2 \beta}{\beta^2}$$

for $I(\theta)$ to be max, we use

$$\frac{dI(\theta)}{d\beta} = 0 = \frac{\beta^2 (2 \sin \beta \cos \beta) - \sin^2 \beta (\beta^2)}{\beta^4} = 0$$

$\Rightarrow$

$$\boxed{\tan \beta = \beta}$$

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$$\text{Where } \beta = \frac{\pi b \sin \theta}{\lambda}$$

⑦ Illumination pattern due to Single Slit Diffraction:

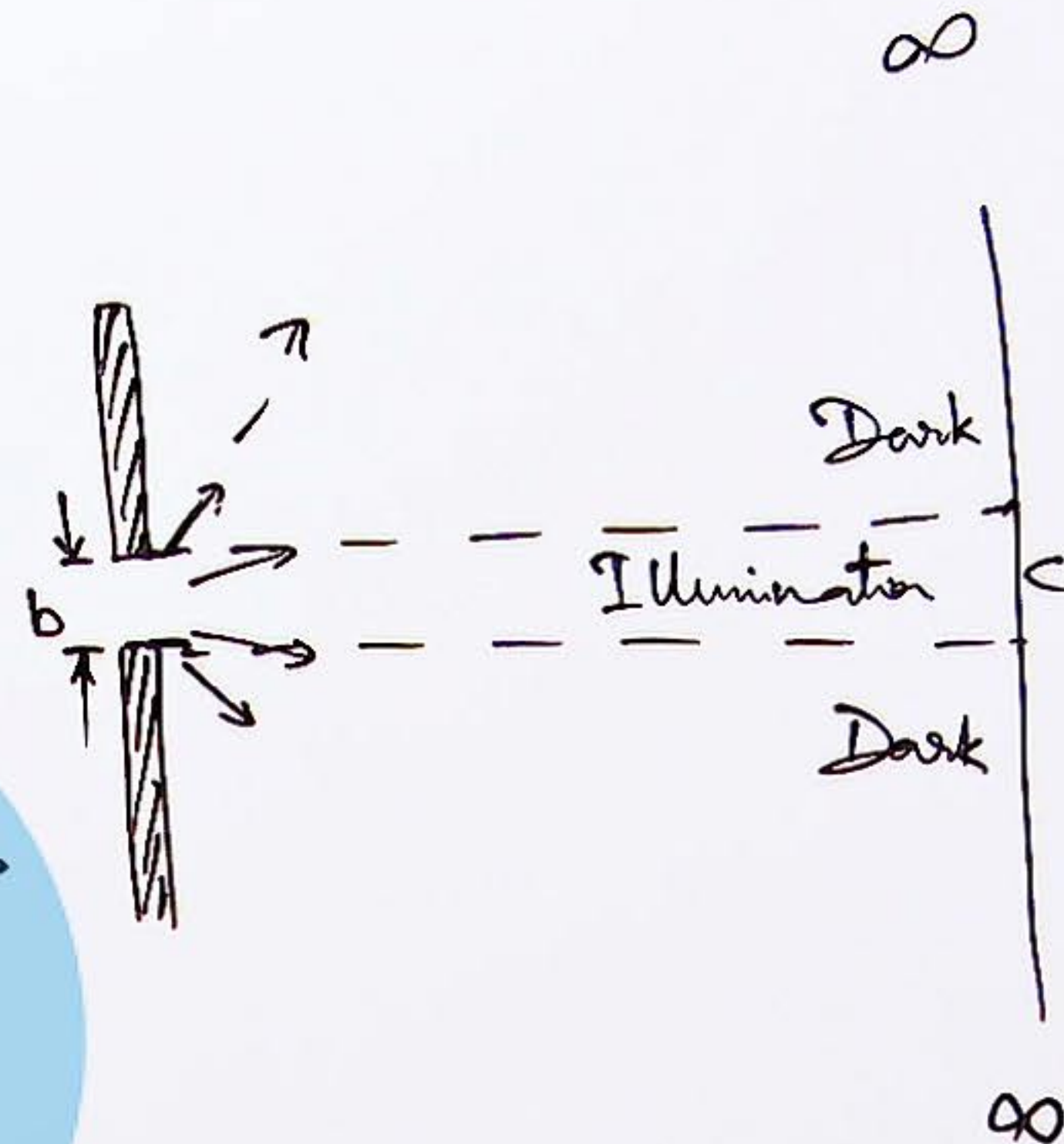
First minima at $\theta_1 = \sin^{-1}\left(\frac{\lambda}{b}\right)$

Case-1: If $b \gg \lambda \Rightarrow \theta_1 \approx 0$
 $\Rightarrow$ Almost no Diffraction

Case-2: If $b > \lambda \Rightarrow \theta_1 \rightarrow$ finite values.
 $\Rightarrow$ Prominent diffraction

Case-3: If $b = \lambda \Rightarrow \theta_1 = \pi/2$
 $\Rightarrow$ No dark fringes are seen.

Case-4: If $b < \lambda \rightarrow$ Same illumination on whole screen decreasing on either side of central pt.



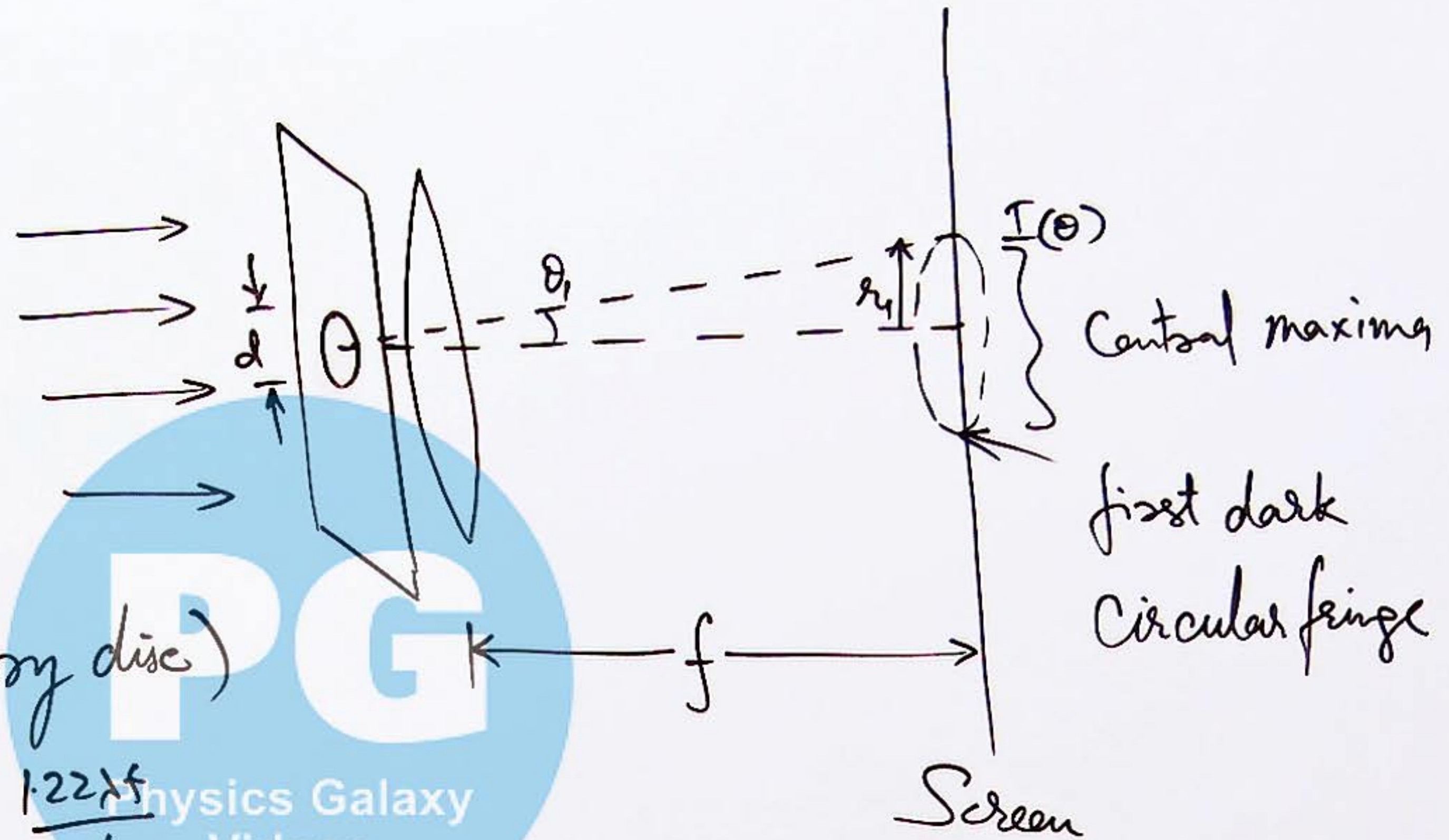
⑧ Diffraction by a Circular Aperture :

angular
first minima on screen

$$\theta_1 = \frac{1.22\lambda}{d}$$

Radius of Central max (Airy disc)

$$r_1 = f\theta = \frac{1.22\lambda f}{d}$$



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